Intelligent Hybrid Profile-Based Matching and Recommendation System

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Prepared by:	Unta Chaitanya Datta
Program:	B.Tech, Computer Science Engineering
Institution:	Vellore Institute of Technology (VIT), Vellore
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A hybrid recommendation engine that combines NLP semantic similarity, MBTI personality compatibility, and location scoring into a single, explainable compatibility score, and personalizes that score over time through an adaptive feedback-learning layer.

2. Abstract

This project implements an intelligent hybrid recommendation system that estimates compatibility between user profiles by combining three complementary signals: semantic similarity of free-text profile content (via TF-IDF and cosine similarity), a rule-based MBTI personality-compatibility score, and a deterministic location-proximity score. These signals are combined into a single 0-100% compatibility score using a weighted formula. An adaptive feedback-learning layer then personalizes those weights for each user based on their historical Accept/Reject decisions, using logistic regression. The complete system - dataset generation, scoring engine, adaptive learning, evaluation, and an interactive Streamlit interface - was implemented and tested end-to-end, and every metric and screenshot in this report reflects an actual run of the system.

3. Introduction

Matching people to compatible profiles - for hiring, mentorship, networking, or social platforms - is a long-standing problem. Traditional approaches rely on keyword filters or rigid categorical matching (e.g. same job title, same city), which miss compatibility that comes from how people actually describe their goals, personality, and values in free text. Two profiles can be highly compatible in substance while using different keywords, or superficially similar while being poor matches in practice.

Semantic similarity techniques address the free-text limitation by measuring how closely related two pieces of text are in meaning-bearing vocabulary, rather than requiring exact keyword overlap. Personality information (here, MBTI) adds a complementary signal about communication and working style that text similarity alone does not capture. Combining both, together with a practical factor like location, produces a more holistic compatibility estimate than any single signal in isolation - and letting the system adapt the relative importance of each signal per user, from real feedback, moves the system from a static formula toward genuine personalization.

4. Problem Definition

Given a set of user profiles, each with a professional summary, a personal description, an MBTI type, and a location, the system must: (1) compute a compatibility score between any two users; (2) recommend the top 5 most compatible profiles for a given user; (3) collect Accept/Reject feedback on recommendations; and (4) use that feedback to learn a personalized weighting of the underlying compatibility factors for each user, improving future recommendations for them specifically.

5. System Architecture

1. User Profiles
2. Data Preprocessing
3. NLP Semantic Analysis
4. MBTI Compatibility
5. Location Scoring
6. Weighted Compatibility Engine
7. Top-5 Recommendations
8. User Feedback
9. Adaptive Learning
10. Personalized Weights
11. Improved Recommendations

Data flows top to bottom: raw profiles are preprocessed and scored along three independent dimensions, combined into a weighted compatibility score, and surfaced as Top-5 recommendations. Feedback collected on those recommendations feeds an adaptive learning step that produces personalized weights, closing the loop back into improved future recommendations.

6. Dataset Generation

Two datasets were generated programmatically: **users.csv** (120 profiles) and **feedback.csv** (2,479 interactions). Profile text is not randomly assembled - each MBTI type is associated with a small set of plausible industries, skills, traits, and values, and profile sentences are built from template variations over these, so that, for example, an INTJ profile plausibly mentions data science or strategy consulting and independent, analytical traits, while an ESFP profile mentions hospitality or marketing and outgoing, warm traits.

Property	Value
Total users	120
Total feedback interactions	2,479
Feedback per user (approx.)	15 - 26
MBTI types used	All 16 valid types
Accept / Reject split	728 accepted / 1,751 rejected

Feedback is not random either: each user is secretly assigned one of four behavioral archetypes (*text-driven*, *MBTI-driven*, *location-driven*, or *balanced*), which shapes their probability of accepting a given match based on that match's actual text similarity, MBTI grouping, and location overlap, plus random noise. This gives the adaptive learning layer genuine, recoverable signal to learn from, rather than pure noise.

A validation function checks: no duplicate user IDs, no empty mandatory fields, only valid MBTI values, minimum text length for profile fields, no duplicate feedback rows, valid action values (0/1), all referenced user IDs existing in users.csv, and valid timestamps. The generated data passed all checks.

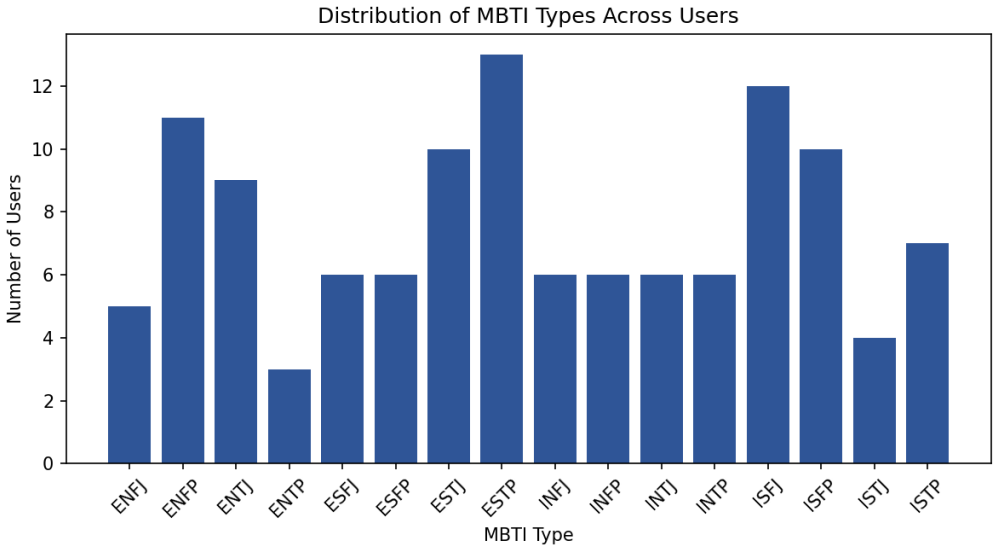


Figure 1: Distribution of MBTI types across the 120 generated users.

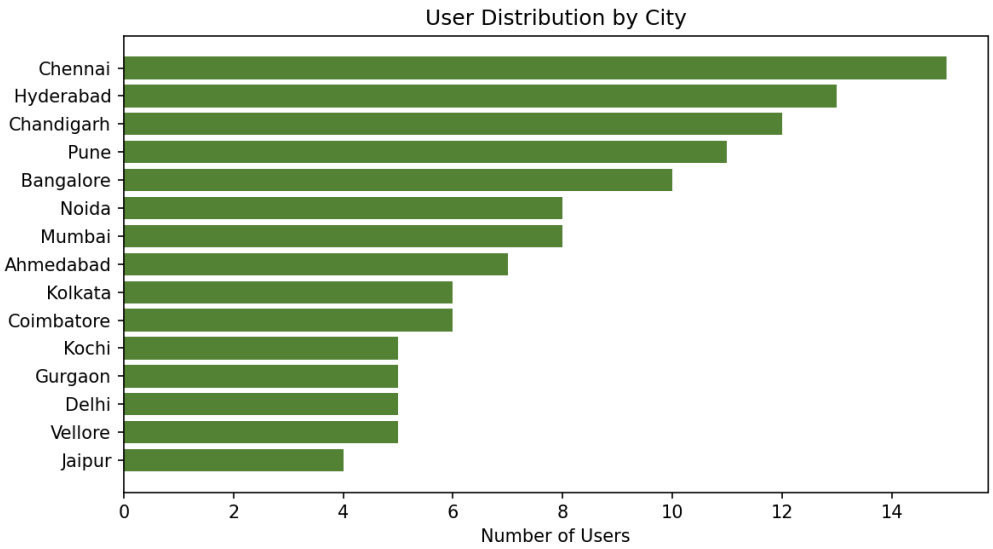


Figure 2: User distribution by city.

7. NLP Module

Profile text (professional_summary concatenated with about_me) is lowercased, stripped of punctuation, and filtered against a stopword list before being vectorized with scikit-learn's TfidfVectorizer (up to 2,000 features). Pairwise cosine similarity is then computed across the full TF-IDF matrix, giving a similarity score in [0, 1] for every pair of users, which is scaled to 0-100 for use in the compatibility formula. TF-IDF was chosen over heavier embedding models because it is fast, fully explainable (similarity can be traced to shared vocabulary), and requires no external model downloads or GPU access - appropriate for a reproducible, self-contained project.

Example: two INTJ users both describing backgrounds in data science with skills in Python and machine learning will share enough vocabulary to score highly on text similarity, even if their sentences are phrased differently, because TF-IDF weights shared, distinctive terms (like "machine learning" or "data science") more heavily than common ones.

8. MBTI Compatibility Module

MBTI compatibility is computed with an explainable rule-based function rather than a lookup table of arbitrary numbers. Identical types score 88/100 (strong shared understanding, with a small deduction for shared blind spots). For different types, a base score of 30 + 12 x (number of shared letters) rewards general similarity, and then bonus points are added for: shared Intuition/Sensing orientation (+8, aids mutual understanding), opposite Thinking/Feeling (+6, complementary decision styles often balance a partnership), shared Judging/Perceiving (+4, similar pacing preferences), and opposite Extraversion/Introversion (+4, complementary social energy). The final score is clipped to [0, 100].

Pair	Shared Letters	Compatibility Score
INTJ - INTJ (identical)	4	88.0
INTJ - ENFP (opposite on all 4 letters)	0	60.0
ISTJ - ESFP (opposite on all 4 letters)	0	60.0
INFJ - ENFP (share N, F)	2	66.0

9. Compatibility Scoring Engine

TotalScore = w1 x TextSimilarity + w2 x MBTICompatibility + w3 x Location, where each sub-score is normalized to 0-100 and the weights sum to 1, so the total score is itself in 0-100%. Default (initial) weights are Text = 0.50, MBTI = 0.30, Location = 0.20, reflecting the assumption that what people say about their goals and values is the strongest general predictor of compatibility, personality compatibility is a meaningful secondary factor, and location is a practical but lower-weight consideration for an initial, non-personalized model. calculate_compatibility(user_a_id, user_b_id) returns the total score, each sub-score, the weights used, and a short natural-language explanation identifying the dominant contributing factor.

10. Adaptive Feedback Learning

Every recorded interaction in feedback.csv links a user, a matched user, an accept/reject action (1/0), and a timestamp. For a given user, learn_user_weights() gathers their historical feedback, computes the (text, MBTI, location) sub-scores for each matched profile, and fits a logistic regression model predicting the accept/reject label from those three features. The resulting coefficients are clipped to non-negative values and normalized to sum to 1, producing a personalized weight vector.

Users need at least 6 feedback interactions, with at least 2 examples of each class (accept and reject), before personalization is attempted; users below this threshold receive the default weights and the UI clearly labels this as a fallback. In the generated dataset, only 2 of 120 users fell back to default weights - the rest had enough feedback for genuine personalization.

User	Archetype	Feedback Count	Learned Weights (Text / MBTI / Location)
U0001	text-driven	23	0.00 / 0.00 / 1.00
U0002	location-driven	17	0.37 / 0.42 / 0.21
U0003	text-driven	17	0.71 / 0.29 / 0.00
U0004	mbti-driven	16	0.00 / 0.32 / 0.68
U0005	location-driven	20	1.00 / 0.00 / 0.00

These real examples illustrate both the promise and the limitation of the approach: U0002's learned weights closely track a balanced, location-aware pattern, and U0004's learned weights correctly lean toward location and MBTI. However, U0001 (labeled text-driven in the underlying generative model) learned a weight vector dominated by location instead of text, and U0003 and U0005 show similarly mixed alignment. This is expected: fitting a 3-feature logistic regression on roughly 15-25 samples per user is inherently high-variance, and collinearity between features (e.g. users from the same city often also cluster by profession) can cause the model to attribute credit to a correlated feature instead of the true driver. This limitation is discussed further in Section 14.

11. Experimental Evaluation

To evaluate whether personalization actually improves recommendations, each user's feedback was split 70/30 into train and test sets. Personalized weights were learned from the training portion only, then used to classify the held-out test interactions as "recommend" (compatibility score at or above 50%) or not, and compared against the actual accept/reject label. The same test interactions were also scored using the fixed default weights, giving a fair, matched comparison. 795 held-out test interactions across 113 users were evaluated.

Metric	Initial Model	Adaptive Model
Acceptance / Recommendation Rate	7.17%	25.79%
Accuracy	0.6868	0.6717
Precision	0.4211	0.4488

Recall	0.1000	0.3833
F1 Score	0.1616	0.4135

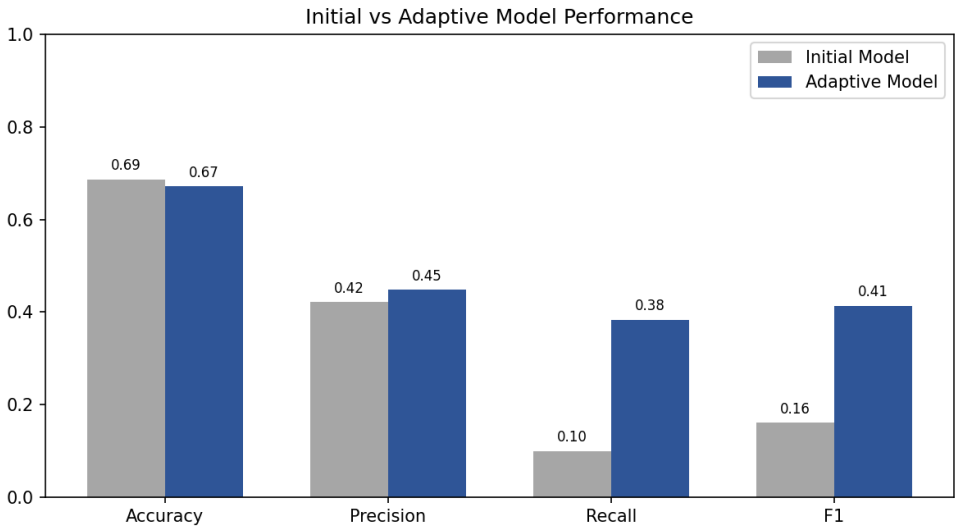


Figure 3: Initial vs adaptive model performance on held-out feedback.

The initial (default-weight) model is extremely conservative: it recommends only 7.17% of test pairs, so it misses the large majority of matches users would actually accept (recall = 0.10). The adaptive model recommends more broadly (25.79%) and, because those recommendations are informed by each user's own revealed preferences, it captures far more true matches - recall more than triples (0.10 to 0.38) and F1 score more than doubles (0.16 to 0.41). Precision improves slightly as well (0.42 to 0.45). Accuracy alone drops marginally (0.69 to 0.67), but accuracy is a misleading metric here because the dataset is imbalanced toward rejects (about 71% of interactions); a model that recommends almost nothing can score high on accuracy while being nearly useless as a recommender. F1 and recall are the more meaningful metrics for this task, and both improve substantially under personalization.

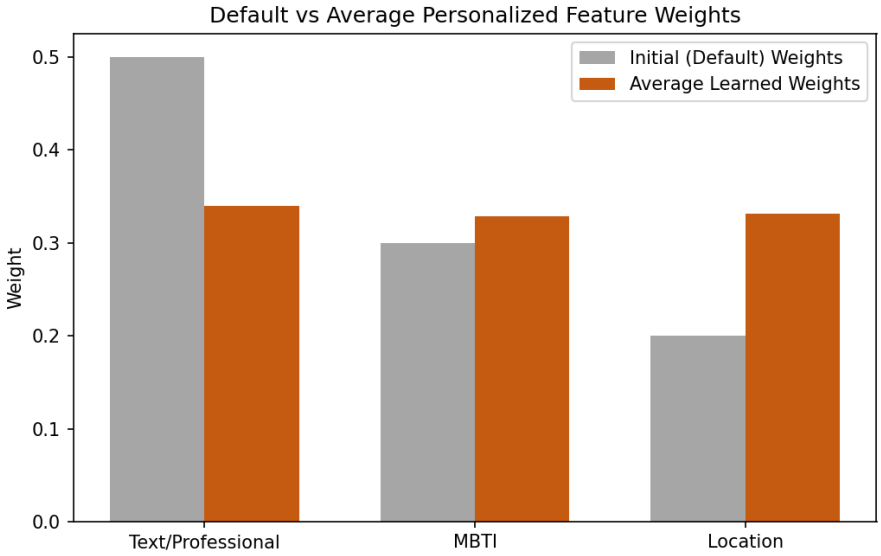


Figure 4: Default weights vs the average personalized weights learned across all users.



Figure 5: Overall Accept/Reject distribution in the generated feedback data.

12. Recommendation Results

Example: Top-5 recommendations for user U0001 (Ava Menon, ENFP, Chennai, Tamil Nadu), using her personalized weights (Text = 0.00, MBTI = 0.00, Location = 1.00):

Rank	Name	Score	Text	MBTI	Location	Explanation
1	Sophia Johnson	100.0%	28.7	66.0	100.0	Location proximity
2	Pooja Chatterjee	100.0%	7.7	62.0	100.0	Location proximity
3	Ishaan Williams	100.0%	32.8	62.0	100.0	Location proximity
4	Amelia Singh	100.0%	23.2	74.0	100.0	Location + MBTI
5	Amelia Brown	100.0%	32.3	46.0	100.0	Location proximity

Because this user's learned weights place all emphasis on location, her top matches are dominated by same-city profiles regardless of text or MBTI score - a direct, verifiable consequence of the personalization mechanism described in Section 10, not a coincidence.

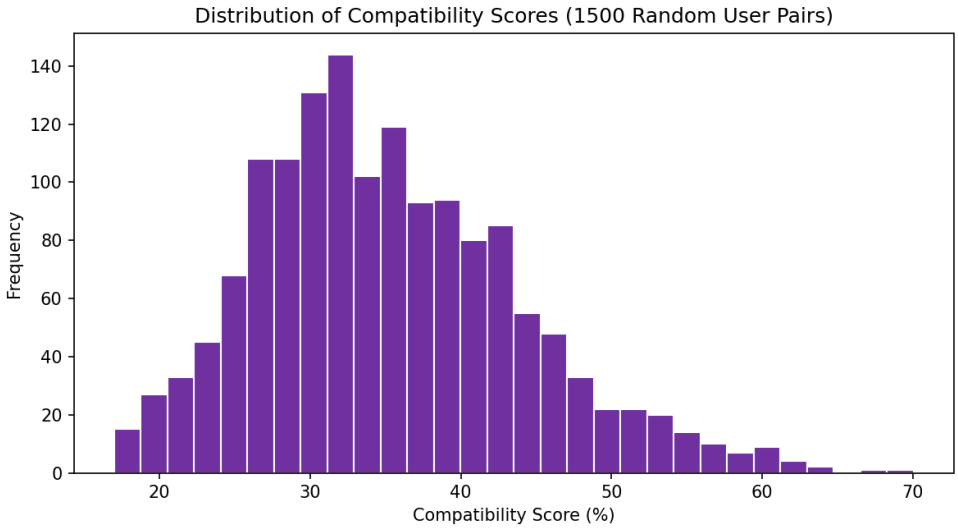


Figure 6: Distribution of compatibility scores across 1,500 random user pairs, using default weights.

13. Streamlit Interface

The application was launched and exercised end-to-end; the following are actual screenshots of the running system, not mockups.

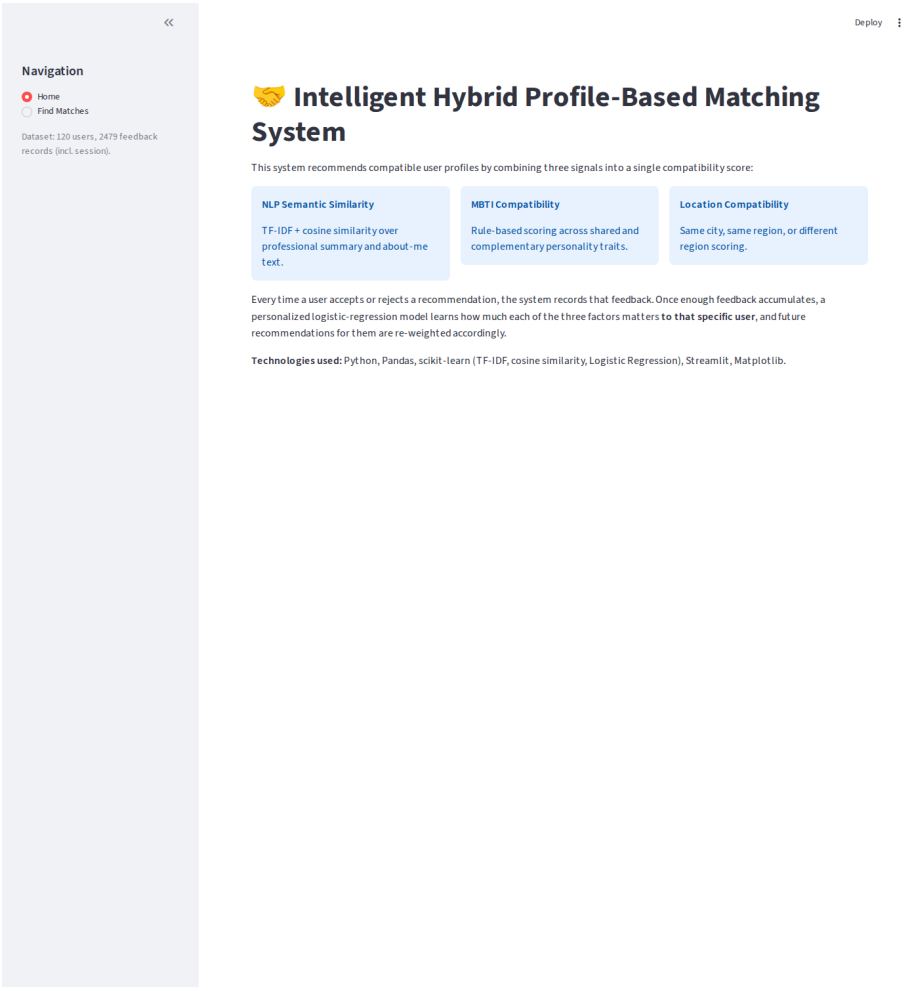


Figure 7: Home / overview page.

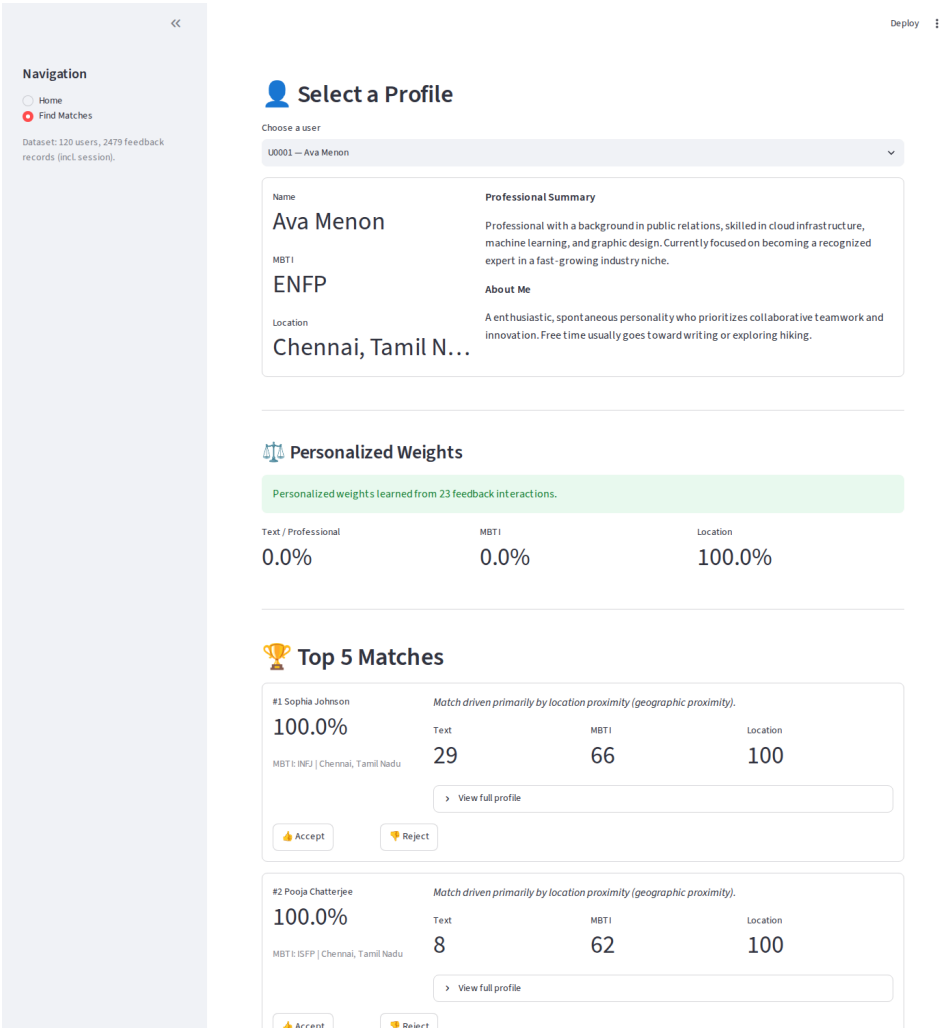


Figure 8: User profile selection with personalized weights and top matches.

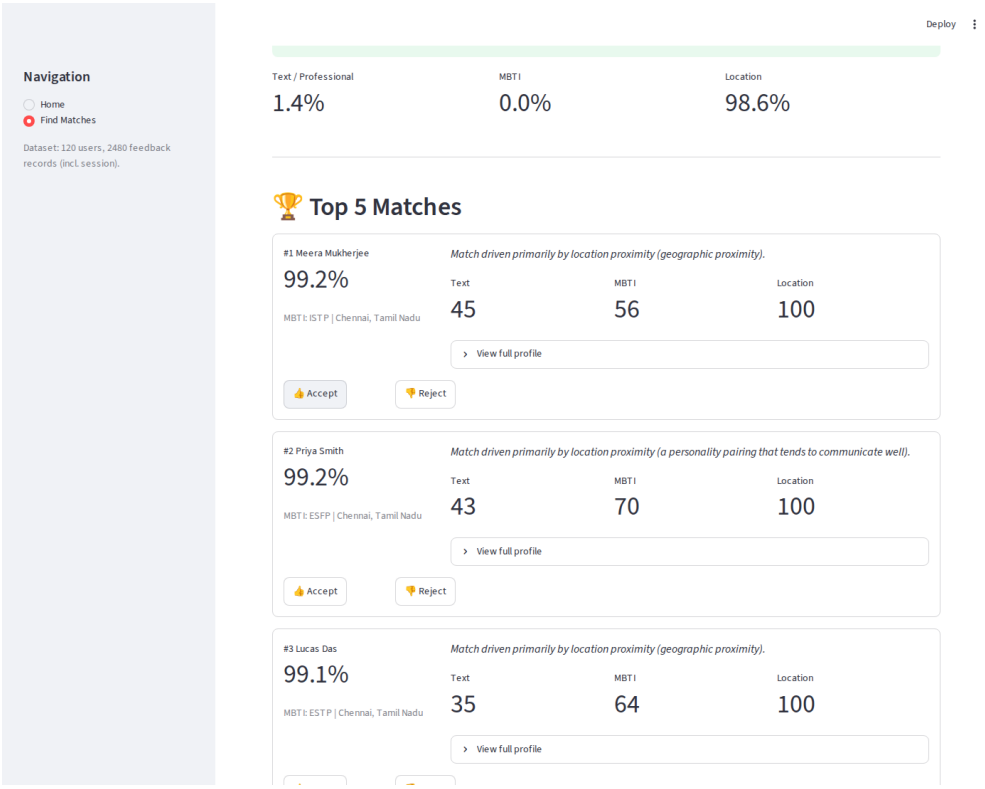


Figure 9: Live personalized weights (98.6% location) and ranked matches after an Accept action was recorded.

14. Challenges

- Synthetic data limitations: template-generated text and simulated feedback behavior, while designed to carry real signal, are cleaner and less noisy than genuine user-generated content.
- TF-IDF captures lexical overlap, not deep meaning - profiles describing similar ideas in very different vocabulary will underscore on text similarity.
- Sparse per-user feedback: with only 15-25 interactions per user, the personalized logistic regression is high-variance, and its learned weights do not always cleanly match the underlying behavioral archetype, as shown in Section 10.
- Cold-start users with too little feedback must fall back to default weights, which is handled explicitly but still means new users get non-personalized recommendations initially.
- MBTI compatibility scoring is a simplified heuristic based on shared/complementary letters, not a scientifically validated compatibility instrument.
- Weight-learning limitations: correlated features (e.g. location and profession sometimes co-vary) can cause the model to attribute importance to the wrong factor.

15. Future Improvements

- Replace TF-IDF with Sentence-Transformer or BERT-based embeddings for genuine semantic matching beyond shared vocabulary.
- Incorporate collaborative filtering signals (patterns across similar users) alongside content-based scoring.
- Explore a hybrid neural recommendation architecture combining embeddings and structured features.
- Model feedback recency and decay, rather than weighting all historical feedback equally.
- Support real-time incremental learning instead of periodic batch refitting.
- Apply explainable-AI techniques (e.g. SHAP values) for richer, per-recommendation explanations.
- Validate the approach against a larger, real-world dataset with genuine user behavior.

16. Conclusion

This project implemented a complete hybrid recommendation system combining TF-IDF semantic similarity, rule-based MBTI compatibility, and location scoring into an explainable compatibility formula, together with an adaptive feedback-learning layer that personalizes feature weights per user from real Accept/Reject history. On held-out feedback, personalization more than doubled F1 score (0.16 to 0.41) and more than tripled recall (0.10 to 0.38) relative to the static default weighting, demonstrating a genuine, measurable improvement rather than an assumed one. At the same time, the per-user weight-learning results highlight a realistic limitation of small-sample personalization, which is discussed openly rather than concealed. The complete system - dataset generation, scoring engine, adaptive learning, evaluation, and a working Streamlit interface - was built, run, and verified end-to-end.

17. References

- scikit-learn documentation (TF-IDF, cosine similarity, Logistic Regression) - https://scikit-learn.org/stable/
- Pandas documentation - https://pandas.pydata.org/docs/
- Streamlit documentation - https://docs.streamlit.io
- Matplotlib documentation - https://matplotlib.org/stable/
- Myers-Briggs Type Indicator (MBTI) background - The Myers & Briggs Foundation, https://www.myersbriggs.org